Covalent and metallic bonding, and electrolysis

20 questions: 7 multiple choice, 13 structured.

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1 · Part 3

Paper 1 & 2 | Covalent bonding | [1]

Which statement describes a covalent bond?

A. the transfer of electrons from a metal atom to a non-metal atom

B. a pair of electrons shared between two atoms

C. the attraction between oppositely charged ions

D. the force of attraction between two separate molecules

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2 · Part 3

Paper 1 & 2 | Covalent bonding | [1]

Which molecule contains a double covalent bond?

A. water, H_2O

B. ammonia, NH_3

C. nitrogen, N_2

D. oxygen, O_2

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3 · Part 3

Paper 1 & 2 | Covalent bonding | [1]

The table shows some properties of four substances.

Which substance is most likely to have a giant covalent structure?

	Melting point in °C	Conducts electricity when solid?	Conducts electricity when liquid?
A	1600	no	no
B	801	no	yes
C	−23	no	no
D	98	yes	yes

4 · Part 3

Paper 2 only | Metallic bonding | [1]

Gold can be hammered into very thin sheets without breaking.

Why can gold be hammered into thin sheets?

A Gold atoms are held together by strong covalent bonds in all directions.

B The gold ions are free to move through the metal.

C The layers of gold atoms can slide over each other without the metallic bonding being broken.

D There are only weak forces between gold atoms.

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5 · Part 3

Paper 2 only | Electrolysis | [1]

Molten lead(II) bromide is electrolysed using inert electrodes.

Which row shows the ion attracted to the cathode and the product formed at the cathode?

	Ion attracted to the cathode	Product formed at the cathode
A	Pb^{2+}	lead
B	Br^{-}	bromine
C	Pb^{2+}	bromine
D	Br^{-}	lead

6 · Part 3

Paper 2 only | Electrolysis | [1]

Concentrated aqueous potassium bromide is electrolysed using inert electrodes.

Which row shows the products at the cathode and at the anode?

	Product at the cathode	Product at the anode
A	potassium	bromine
B	bromine	hydrogen
C	hydrogen	bromine
D	hydrogen	oxygen

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7 · Part 3

Paper 2 only

Electrolysis

[1]

Aqueous silver nitrate is electrolysed using inert electrodes. Silver is deposited on the cathode.

Which row shows the half-equation for the reaction at the cathode and the type of reaction?

	Half-equation at the cathode	Type of reaction
A	$Ag^+ + e^- \rightarrow Ag$	oxidation
B	$Ag^+ + e^- \rightarrow Ag$	reduction
C	$Ag \rightarrow Ag^+ + e^-$	oxidation
D	$Ag \rightarrow Ag^+ + e^-$	reduction

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8 · Part 3

Paper 1 & 2 | Covalent bonding | [5]

Methane, CH_4, is a covalent compound. In a methane molecule, each hydrogen atom is joined to the carbon atom by a single covalent bond.

(a) State the number of covalent bonds in one molecule of methane. [1]

(b) State the total number of electrons shared in one molecule of methane. [1]

(c) Explain, in terms of electrostatic attraction, how a covalent bond holds two atoms together. [2]

(d) State whether methane conducts electricity. [1]

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9 · Part 3

Paper 1 & 2 Covalent bonding [6]

The table gives information about five compounds of carbon and hydrogen. All five compounds have simple molecular structures.

Compound	Formula	Relative molecular mass (Mr)	Boiling point in °C
methane	CH_4	16	−162
ethane	C_2H_6	30	−89
propane	C_3H_8	44	−42
butane	C_4H_{10}	58	−1
pentane	C_5H_{12}	72	36

(a) Describe how the boiling point changes as the relative molecular mass increases. [1]

(b) Estimate the boiling point of hexane, C_6H_{14}, which has a relative molecular mass of 86. [1]

(c) Deduce which of the compounds in the table are gases at 25 °C. Give a reason for your answer. [2]

(d) Explain why the boiling point increases as the relative molecular mass increases. [2]

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10 · Part 3

Paper 1 & 2 | Covalent bonding | [6]

Silicon dioxide, SiO_2, and carbon dioxide, CO_2, both contain covalent bonds.

Silicon dioxide melts at 1710 °C. Solid carbon dioxide changes directly into a gas at −78 °C.

(a) Give one reason why silicon dioxide is suitable for lining the inside of a furnace. [1]

(b) Explain, in terms of structure and bonding, why silicon dioxide must be heated to a much higher temperature than carbon dioxide before it changes state. [5]

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11 · Part 3

Paper 1 & 2 | Covalent bonding | [6]

Diamond and graphite are both forms of carbon with giant covalent structures. Graphite is used to make electrodes. Diamond is used on the tips of drills for cutting rock.

(a) Explain why graphite conducts electricity. [2]

(b) Explain, in terms of structure and bonding, why diamond is very hard but graphite is soft and slippery. [4]

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12 · Part 3

Paper 1 & 2 | Covalent bonding | [5]

C_{60} fullerene is another form of carbon. It consists of separate molecules. In each molecule, 60 carbon atoms are joined together to form a hollow ball shape.

(a) State the type of bond between the carbon atoms within a C_{60} molecule. [1]

(b) Explain why C_{60} fullerene is much softer than diamond. [3]

(c) Graphite and C_{60} fullerene are both forms of carbon. State one similarity, other than the type of bond, in how each carbon atom is bonded in graphite and in C_{60} fullerene. [1]

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13 · Part 3

Paper 1 & 2 | Covalent bonding | [6]

Iodine is a dark grey solid made of I_2 molecules. When iodine crystals are warmed gently, a purple vapour forms. The vapour contains I_2 molecules, not separate iodine atoms.

(a) State the type of structure that iodine has. [1]

(b) Explain why only gentle warming is needed to turn solid iodine into a vapour. [2]

(c) The vapour contains I_2 molecules. Explain what this shows about the strength of the covalent bond in I_2 compared with the forces between I_2 molecules. [2]

(d) State whether solid iodine conducts electricity. [1]

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14 · Part 3

Paper 2 only

Metallic bonding

[5]

Magnesium is a metal. It melts at 650 °C.

(a) Describe the structure and bonding in solid magnesium. [3]

(b) Explain why magnesium has a high melting point. [2]

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15 · Part 3

Paper 2 only Metallic bonding [6]

Copper is used to make electrical wires. Copper(II) chloride, $CuCl_2$, is an ionic compound.

(a) Explain why solid copper conducts electricity. [2]

(b) Solid copper(II) chloride does not conduct electricity. Explain why. [2]

(c) When an electric current passes through molten copper(II) chloride, the compound is decomposed. When an electric current passes through a copper wire, the copper does not change. Explain this difference. [2]

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16 · Part 3

Paper 2 only

Electrolysis

[6]

Lena tests two solutions for electrical conductivity. She dips two graphite electrodes into each solution. The electrodes are connected to a battery and a lamp. Glucose, $C_6H_{12}O_6$, is a covalent compound.

Solution	Does the lamp light?	Observation at the electrodes
sodium chloride solution	yes	bubbles of gas form
glucose solution	no	no change

(a) State what is meant by the term anion. [1]

(b) Explain why sodium chloride solution conducts electricity. [2]

(c) Explain why glucose solution does not conduct electricity. [2]

(d) Suggest why bubbles of gas form at the electrodes only in the sodium chloride solution. [1]

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17 · Part 3

Paper 2 only

Electrolysis

[6]

Molten zinc chloride, $ZnCl_2$, is electrolysed in a fume cupboard using inert graphite electrodes. Zinc forms at the cathode and chlorine forms at the anode.

(a) Give the formula of the cation and the formula of the anion in zinc chloride. [2]

(b) Give the ionic half-equations for the reactions at the cathode and at the anode. [2]

(c) Explain why the reaction at the anode is described as oxidation. [1]

(d) Suggest why the electrolysis is carried out in a fume cupboard. [1]

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18 · Part 3

Paper 2 only

Electrolysis

[6]

Concentrated aqueous sodium chloride is electrolysed using graphite electrodes. Each electrode is covered by an upside-down test tube filled with the solution, so that any gas produced is collected.

(a) Name the gas produced at the cathode and the gas produced at the anode. [2]

(b) Describe a test to identify the gas produced at the anode. Give the result of the test. [2]

(c) Explain why hydrogen, and not sodium, is produced at the cathode. [2]

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19 · Part 3

Paper 2 only

Electrolysis

[7]

Tomas electrolyses dilute sulfuric acid using inert electrodes. He collects the gas from each electrode in a separate measuring cylinder. After 8 minutes, he has collected 40 cm^3 of gas at the cathode and 19 cm^3 of gas at the anode.

(a) Name the gas collected at the cathode and the gas collected at the anode. [2]

(b) Give the ionic half-equation for the reaction at the anode. [2]

(c) Use the half-equations for the reactions at both electrodes to explain why the volume of gas at the cathode is about twice the volume of gas at the anode. [2]

(d) The volume of gas at the anode is slightly less than half the volume at the cathode. Suggest one reason for this. [1]

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20 · Part 3

Paper 2 only

Electrolysis

[7]

Arjun investigates the electrolysis of copper(II) sulfate solution using inert electrodes. Copper is deposited on the cathode and bubbles of gas form at the anode. Every 5 minutes he removes the cathode, washes it, dries it and weighs it. The table shows his results.

Time in minutes	Increase in mass of cathode in g
0	0.00
5	0.11
10	0.22
15	0.27
20	0.44
25	0.55

(a) State the independent variable and one variable that Arjun should control. [2]

(b) Identify the anomalous result. [1]

(c) Use the results to predict the increase in mass of the cathode after 40 minutes. Show your working. [2]

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(d) The blue colour of the copper(II) sulfate solution becomes paler during the experiment. Explain why. [2]

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